

Using Sets and Venn Diagrams
to Organize Real-World Data
Math 140 – Math for Elementary Teachers

Name: _____

Date: _____

About the Data

In 2020, several large wildfires burned areas of California. Researchers wanted to learn more about what happens in different environments after a wildfire. They collected information from different locations. Some locations had **burned** in the fires, while others had **not burned**. The locations also had different types of habitats, such as grasslands, forests, and areas with shrubs.

Researchers collected information during **2020 and 2021**. Each row in our Excel file represents one observation from the study.

Our Focus

For this activity, we are **not** studying the science of wildfires. Instead, we will use this real information to practice **organizing data into sets**.

Words to Know

Habitat	The type of natural environment where plants and animals live.
Burned	An area affected by the wildfire.
Unburned	An area that was not affected by the wildfire.
Observation	One row of information in our dataset.

Part 1: Start by Hand

Before using Excel, let’s organize a small amount of data by hand.

For this first activity, define

$A = \{\text{observations from burned sites}\}$

$B = \{\text{observations from grassland sites}\}$

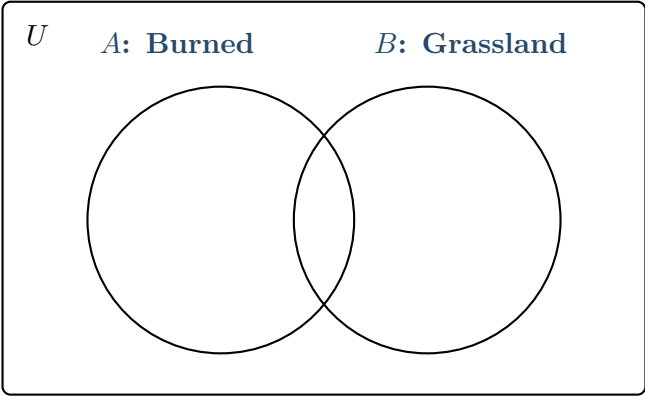
Consider the sample data below.

Observation	Burn Status	Habitat	Year
1	Burned	Grassland	2021
2	Unburned	Grassland	2020
3	Burned	Scrub	2021
4	Unburned	Oak Woodland	2020
5	Burned	Grassland	2020
6	Unburned	Grassland	2021
7	Burned	Oak Woodland	2021
8	Burned	Grassland	2021
9	Unburned	Scrub	2020
10	Burned	Scrub	2020

1. List the observation numbers that belong to set A .
2. List the observation numbers that belong to set B .
3. Which observations belong to $A \cap B$?
4. Which observations belong to neither A nor B ?

Complete the Venn Diagram

Place the **observation numbers** in the correct regions.



Part 2: Working with the Full Dataset in Excel

The complete dataset contains **123 observations**.
Sorting 123 observations by hand would take a long time. Excel can help us organize and count the information.

Open the file:
Math140_Sets_Data.xlsx

The spreadsheet contains four columns:

Column	What It Tells Us
Observation	Identifies each observation.
Burn Status	Tells whether the site was burned or unburned.
Habitat	Tells the type of habitat.
Year	Tells whether the observation was from 2020 or 2021.

Using a Filter in Excel

Each column has a small filter arrow.

To filter the data:

1. Click the filter arrow at the top of the column.
2. Uncheck **Select All**.
3. Check the category you want to see.
4. Click **OK**.

Only rows that match your choice will now be shown.

Important: Clear your previous filters before starting a new count unless you want to use more than one filter at the same time.

Part 3: Two Sets Using the Full Dataset

For this investigation, define

$$A = \{\text{observations from burned sites}\}$$

$$B = \{\text{observations from grassland sites}\}$$

There are **123 total observations**, so

$$n(U) = 123$$

Step 1: Find the Number of Burned Observations

Use the **Burn Status** column in Excel.

1. Click the filter arrow for **Burn Status**.
2. Select only **Burned**.
3. Determine how many observations remain.
4. Record your answer below.

$$n(A) = \underline{\hspace{2cm}}$$

Checkpoint: You should have found $n(A) = 72$.
If you did, clear the filter before continuing.

Step 2: Find the Number of Grassland Observations

Now use the **Habitat** column.

- 1. Click the filter arrow for **Habitat**.
- 2. Select only **Grassland**.
- 3. Determine how many observations remain.
- 4. Record your answer below.

$n(B) =$ _____

Clear the filter before continuing.

Step 3: Find the Intersection

Now we want observations that are **both Burned and Grassland**.

This represents

$A \cap B$

Use two filters at the same time:

- 1. Filter **Burn Status** to show only **Burned**.
- 2. Keep that filter on.
- 3. Filter **Habitat** to show only **Grassland**.
- 4. Determine how many observations satisfy both conditions.

$n(A \cap B) =$ _____

Clear both filters when you are finished.

Step 4: Organize Your Results

Record the information you found.

Set	Count
Total observations, $n(U)$	123
Burned observations, $n(A)$	_____
Grassland observations, $n(B)$	_____
Burned and Grassland, $n(A \cap B)$	_____

Step 5: Do the Mathematics by Hand

Excel gave us the counts. Now use those counts to complete the Venn diagram **without using Excel**.

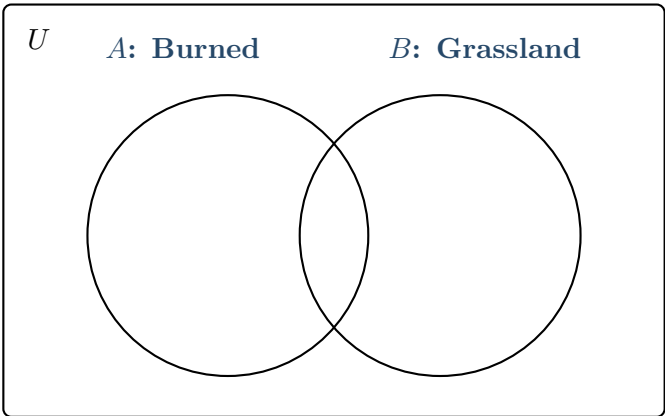
Remember: Start with the intersection.

1. Write $n(A \cap B)$ in the overlap.

2. Determine how many observations are **Burned but not Grassland**.

3. Determine how many observations are **Grassland but not Burned**.

4. Determine how many observations are **neither Burned nor Grassland**.



Step 6: Use Your Venn Diagram

Use your completed Venn diagram to answer the following.

1. Find $n(A \cap B)$.

2. Find $n(A \cup B)$.

3. Find $n(A')$.

4. Find $n(B')$.

5. Find $n(A \cup B')$.

6. Describe $A \cap B$ in words.

7. Describe $A \cup B'$ in words.
- _____

Part 4: Adding a Third Set

Now we will add another characteristic:

$C = \{\text{observations collected in 2021}\}$

Our three sets are now

$A = \text{Burned} \quad B = \text{Grassland} \quad C = \text{2021}$

An observation may belong to **one, two, all three, or none** of these sets.

Use Excel to Find the Following

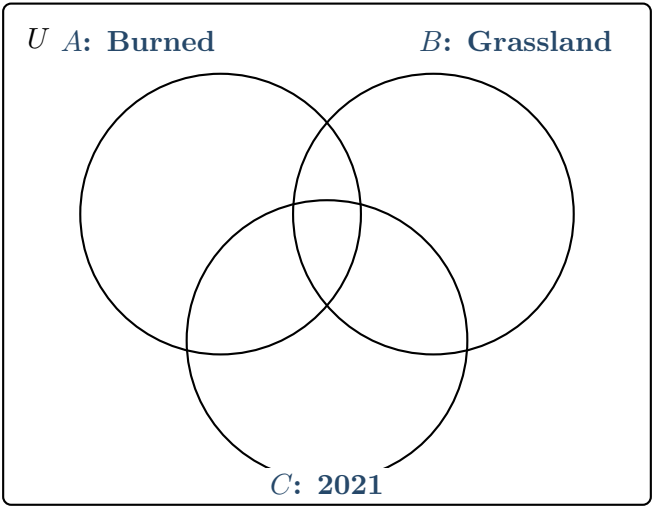
Use the filters in Excel to determine each count.

Set or Intersection	Count
$n(A)$: Burned	_____
$n(B)$: Grassland	_____
$n(C)$: 2021	_____
$n(A \cap B)$: Burned and Grassland	_____
$n(A \cap C)$: Burned and 2021	_____
$n(B \cap C)$: Grassland and 2021	_____
$n(A \cap B \cap C)$: Burned, Grassland, and 2021	_____

Build the Venn Diagram by Hand

Start with the center.

First place $n(A \cap B \cap C)$ in the region shared by all three sets. Then work outward to determine the other regions.



Interpret Your Diagram

1. How many observations belong to all three sets? _____
2. How many observations are Burned and Grassland but are **not** from 2021?
3. How many observations are Burned and from 2021 but are **not** Grassland?
4. How many observations belong to none of the three sets? _____
5. Describe $A \cap B \cap C$ in words.

Part 5: Try It Without Excel

On an exam, you may be asked to construct a Venn diagram **without Excel**.

Suppose a survey of **40 future elementary teachers** found:

- 24 enjoy teaching mathematics.
- 18 enjoy teaching science.
- 10 enjoy teaching both mathematics and science.

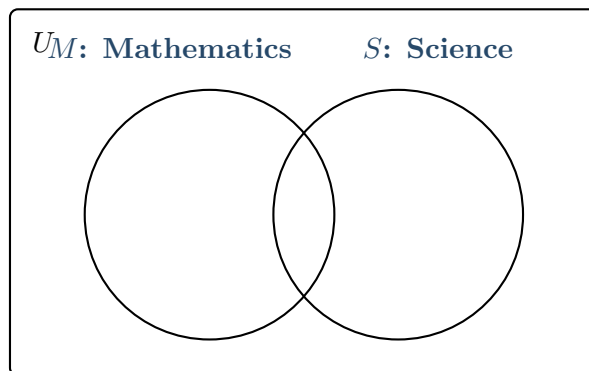
Let

$$M = \{\text{students who enjoy teaching mathematics}\}$$

and

$$S = \{\text{students who enjoy teaching science}\}.$$

1. Complete the Venn diagram.



2. How many students enjoy only mathematics? _____
3. How many students enjoy only science? _____
4. How many students enjoy neither? _____
5. Find $n(M \cup S)$. _____

Final Reflection

Think about both the mathematics and the use of Excel in this activity.

1. In your own words, explain how a Venn diagram helped you organize the information in the dataset.
2. What was one thing Excel made easier when working with all 123 observations?
3. Why was it still important to construct the Venn diagrams by hand instead of relying only on Excel?
4. Give one example of another real-world situation where sets or Venn diagrams could be useful for organizing information.

After completing this activity, how confident are you in using sets and Venn diagrams with real data?

☐ Not yet confident ☐ A little confident ☐ Confident ☐ Very confident

One thing I understand better now is: